- (1) Neuron basics & communication
  • Dendrites (input) → soma → axon (output).
  • Within-neuron: electrical spike (action potential). Between neurons: chemical synapse.
- (2) Brain vocabulary
  • Gray matter=cell bodies; white matter=axons. Two hemispheres linked by corpus callosum. Cortex has lobes (frontal, parietal, temporal, occipital).
- (3) Localization & perception
  • Different brain areas handle different functions; same input can be perceived differently depending on where/when it's processed.
- (4) Three brain methods & fMRI assumptions/limits
  • EEG (fast, low spatial), fMRI (good spatial, **indirect/slow** BOLD), lesions/stimulation (causal but limited). fMRI assumes blood-oxygen changes track local neural activity.
- (5) Converging evidence
  • Strong claims use multiple methods that point to the same conclusion.
- (1) Hippocampus & memory (how we know)
  • Needed for forming new explicit memories (facts/events): patient cases (e.g., HM), imaging during memory tasks, and lesion evidence.
- (2) Plasticity & LTP
  • Plasticity: the brain changes with use.
  • LTP: repeated activity strengthens synapses—cellular basis for learning.
- (3) Other hippocampal roles & "binding"
  • Spatial maps, context, sequence/time; these support "binding elements of experience in time and space."
- (1) Habitual vs deliberative decisions
  • Habitual: fast/automatic. Deliberative: slower, plan/compare options.
- (2) Reward prediction error (RPE) & dopamine
  • RPE = actual − expected reward. Dopamine signals RPE; basal ganglia support habit learning.
- (3) Deliberation experiment (example)
  • Tasks that force planning/tradeoffs; measure choices, reaction times, neural signals during evaluation.
- (4) Applications
  • Habit change, addiction treatment, decision aids; RL algorithms in AI.
- (1) Everyday vs cosmic scales
  • We live at slow speeds/small energies; the universe spans extreme speeds/sizes.
- (2) Constant c puzzle & Einstein's fix
  • Light's speed is always c; Einstein resolved this by letting space and time adjust (not c).
- (3) Simultaneity
  • Same-time for one observer may not be for a moving observer.
- (4) Time & space at high speed; when to use SR
  • Time dilates, lengths contract: $\gamma=1/\sqrt{(1-(v/c)^2)}$. Use SR when v is a big fraction of c.
- (5) $E=mc^2$ & moving fast
  • Mass–energy equivalence; total energy at speed is γmc^2.
- (1) Newton's gravity limits
  • Works for weak gravity/low speed; not exact near massive bodies or for light.
- (2) GR in plain words
  • Mass/energy curve spacetime; motion follows those curves (geodesics).
- (3) Equivalence principle
  • Free-fall ≈ weightlessness locally; gravity indistinguishable from acceleration.
- (4) Evidence for GR
  • Light bending, gravitational redshift, GPS corrections, Mercury's perihelion shift, gravitational waves.
- (5) Black holes & time
  • Schwarzschild radius $r_s=2GM/c^2$. Near mass, clocks run slower (gravitational time dilation). Scale by mass to estimate r_s.
- (6) Expanding universe & Hubble data
  • Farther galaxies recede faster (redshift ∝ distance). Slope tells expansion rate/history.
- (1) Double-slit outcomes & meaning
  • Pellets: two bands. Waves: interference fringes. Electrons: fringes (wave-like); measuring path kills fringes.
- (2) Probability in QM
  • Outcomes are inherently probabilistic; wavefunction gives probabilities.
- (3) Uncertainty principle
  • Can't know position & momentum precisely at once: $\Delta x \cdot \Delta p \geq \hbar/2$.

For each group: $SE = SD/\sqrt{N}$. ; $df = N - 1$, get t^* (95% two-sided). ; $CI = Mean \pm t^* \times SE$. ; A = total highlighted surface area (cm^2) you estimate from the scale bar; T = cortical thickness (cm); f = fraction of cortex that is neuron somata (e.g., 0.03 for 3%); v_{soma} = volume of one neuron soma (cm^3); Cortical volume under the highlight $V_{cortex} = A \times T$ (cm^3); Total soma volume; $V_{soma} = f \times V_{cortex}$; Neuron count $N = V_{soma} / v_{soma} = (f \times A \times T) / v_{soma}$

Welch t (two groups): compare group averages; Lorentz factor: near light effects; ship proper time = Earth time/γ: onboard time experienced; Mass→energy fraction ($E/(mc^2)$): mass energy efficiency; t_E = distance/speed: compute travel time; $\tau = t_E/\gamma$: relativistic onboard time; fraction = $E/(m \cdot c^2)$: mass energy efficiency.

df	t*
1	12.706
2	4.303
3	3.182
4	2.776
5	2.571
6	2.447
7	2.365
8	2.306
9	2.262
10	2.228
11	2.201
12	2.179
13	2.160
14	2.145
15	2.131
16	2.120
17	2.110
18	2.101
19	2.093
20	2.086
21	2.080
22	2.074
23	2.069
24	2.064
25	2.060
26	2.056
27	2.052
28	2.048
29	2.045
30	2.042
∞	1.960